Simplified Spike-timing Dependent Plasticity Learning Rule of Spiking Neural Networks for Unsupervised Clustering

Hongyu Sun[1,2],Xining Cui[1], Yinjing Guo[1], Anqing Ding[1]
1.College of Electronic Information Engineering,Shandong University of Science and Technology,CHINA
2.College of Electrical Engineering and Automation,Shandong University of Science and Technology,CHINA
shygy@126.com,554127842@qq.com,gyjlwh@163.com,aqd6677@163.com

***Abstract*—Spike-Timing Dependent Plasticity (STDP) is widely believed as a family of learning mechanisms originally postulated in the context of artificial machine learning algorithms, and has been proven successful in training spiking neural networks (SNN). This paper presents a simplified model of STDP learning rule in artificial network based on spiking neurons for unsupervised clustering. Population encoder algorithm is used to convert the continuous real-number data to discrete input spike times. Subsequently, the generated spikes are submitted into a two-layered structure SNN which employed Leaky Integrate and Fire (LIF) neurons as nodes and trained by simplified STDP learning rule. The learning and clustering capabilities of simplified STDP were investigated by applying it to the nonlinearly separable data like the 2-Dimension ring data set and 4-Dimension Fisher iris plant classification problem. The results show that the SNN algorithm can achieve the best accuracy of 100% and 95.33% respectively, which demonstrates that the proposed simplified model of STDP learning rule is a viable training algorithm and the SNN approach is suitable for unsupervised clustering.**

***Keywords*—Spiking Neural Networks;Spike-Timing Dependent Plasticity;Unsupervised Clustering**

I. Introduction

SNN is the third generation of Artificial Neural Networks and is more biologically plausible [1–3]. In the past decades, SNN have been successfully used in many domains such as audiovisual information processing [4, 5], EEG classification [6], Very Large Scale Integration (VLSI) System [7], Data Clustering [1].The time is becoming the most important input. The presence and precise timing of spikes encapsulate have a meaning such as human brain behavior. However, deferent techniques are therefore required to submit a stimulus to the neural network to build the timing spike. The characteristics of these spikes are based on their firing time because of the stereotypical nature of the human brain. Neural networks (NN) as engineering tools Operate on analog quantities (analog input, analog output), SNN More powerful than classic NN Interesting to implement in hardware[8].

In this paper, we propose a simplified model of STDP learning algorithm with the advantage of being less computationally expensive than the standard STDP rule. The proposed STDP learning algorithm was used for data clustering. We adopt a winner-take-all method for the post synaptic neurons. For each input vector, as soon as a post synaptic neuron generates a spike, all the others won't generate a spike. The population encoding is used to convert the real-valued data to discrete input spike times. Subsequently, the generated spikes are submitted into a two-layered structure SNN which employed LIF neurons as nodes and trained by simplified STDP learning rule.The nonlinearly separable data like the 2-Dimension ring data set and 4-Dimension Fisher iris plant classification problem were investigated in experiments. The results show that the SNN algorithm can achieve the highest accuracy of 100% and 95.33% respectively, which demonstrates that the proposed simplified model of STDP learning rule is a viable training algorithm and the SNN approach is suitable for unsupervised clustering.

This paper is organized as follows. Section II introduces the STDP model, its simplified model and SNN architecture. In Section III, we introduce the experiment setup and data clustering. The results and data analysis are also presented. Finally, Section IV draws the conclusion.

II. Methodology

A. STDP

STDP is a temporally asymmetric form of Hebbian learning which adjusts the strength of a connection in response to the time difference between the pre-synaptic and post-synaptic spikes [9]. Repeated pre-synaptic spike arrival a few milliseconds before postsynaptic action potential leads to long-term potentiation (LTP) of the synapses, whereas repeated spike arrival after postsynaptic spikes leads to long-term depression (LTD). Mathematically, the STDP learning function is described by a function of relative timing window $W(\Delta t = t_{pre} - t_{post})$ that determines the fractional change of the synaptic weight in terms of the difference between the arrival time t_{pre} of a pre-synaptic spike and the time t_{post} of generation of a post-synaptic spike. The curve in Fig.1 shows a functional approximation to the fitted correlation, which is called the STDP curve, and is represented by the following equation:

$$W(\Delta t) = \begin{cases} A_+ exp(\frac{\Delta t}{\tau_+}) & \text{if } \Delta t \leq 0 \\ A_- exp(-\frac{\Delta t}{\tau_-}) & \text{if } \Delta t > 0 \end{cases} \quad (1)$$

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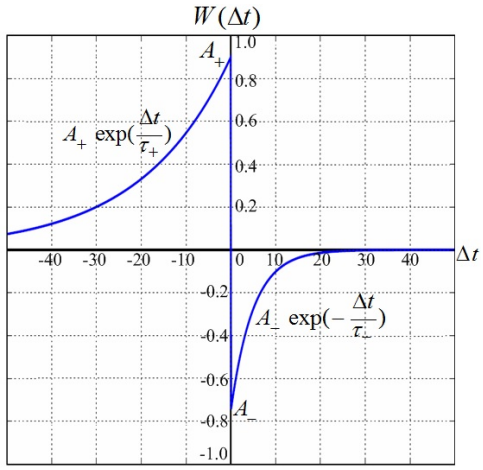

Fig. 1. STDP learning curve.

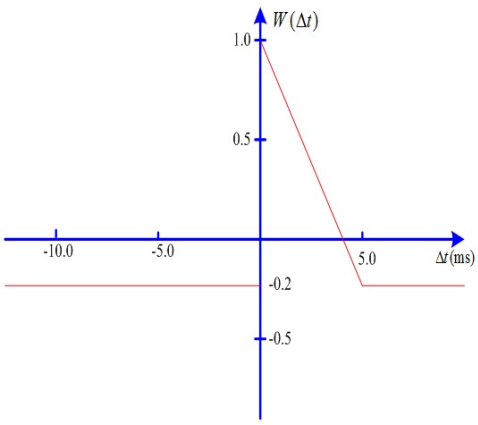

Fig. 2. Simplified STDP learning curve.

where parameters τ_+ and τ_- are the temporal range of the pre-synaptic and post-synaptic time interval, A_+ and A_- delineate the maximum fractions of synaptic modification, at Δt close to zero. Generally speaking, the window W is usually temporally asymmetric ($A_+ \neq A_- \, and \, \tau_+ \neq \tau_-$). Fig. 1 was generated by setting $A_+ = 0.85$, $A_- = -0.75$, $\tau_+ = 20$, and $\tau_- = 5$ in Eq.(1).

B. Simplified STDP

Subsubsection text here. To improve the rate of convergence and the performance of network, we simplify the STDP as the following equation:

$$W(\Delta t) = \begin{cases} 1 - r * \Delta t & \text{if } 0 < \Delta t \leq \frac{1.2}{r} \\ -0.2 & \text{if } \Delta t \leq 0 \, or \, \Delta t > \frac{1.2}{r} \end{cases} \tag{2}$$

where $\Delta t = t_{post} - t_{pre}$ is the difference between the arrival time t_{post} of a post-synaptic spike and the time t_{pre} of a pre-synaptic spike, r indicates the slope of line in Simplified STDP learning curve as showed in Fig.2. The simplified STDP algorithm is summarized with the pseudo-code of Table 1.

TABLE I
THE SIMPLIFIED STDP ALGORITHM

The Pseudo-code of Simplified STDP algorithm
foreach pre-synaptic spike do **foreach** post-synaptic spike do $\Delta t = t_{post} - t_{pre}$; $w = compute_weight_modification(\Delta t)$; $new_weight = updatesynapticweight(current_weight, w)$; **end** **end**
Function description: t_{post} : time of the post-synaptic spike; t_{pre}: time of the pre-synaptic spike; $compute_weight_modification()$:computes the weight modification according to the Simplified STDP curve ; $updateweight()$: updates the synaptic weight according to the current weight and the weight modification previously computed,applying also hard limits on the synaptic weight value, where appropriate;

The weight change is described by Eq.(3). The weights are always limited by $w_{max} \geq w \geq w_{min}$. The weight change rate η controls the weight adaptation speed.

$$w_{new} = \begin{cases} w_{old} + \eta \Delta w(w_{max} - w_{min}), & \text{if } \Delta w > 0 \\ w_{old} + \eta \Delta w(w_{old} - w_{min}), & \text{if } \Delta w \leq 0 \end{cases} \tag{3}$$

C. SNN architecture

To verify this simplified STDP learning rule, a two layered fully connected feed-forward SNN has been used to cluster data. The network structure depicted as in Fig.3. A connection from pre-synaptic neuron i to postsynaptic neuron j consists of m delayed synaptic terminals[10]. Each terminal serves as a sub-connection that is associated with a different delay d^k and weight w_{ij}^k. A spike from neuron i thus generates m delayed spike-response functions $\varepsilon(t - (t_i + d^k))$, the sum of which generates the membrane-potential in neuron j. When the membrane-potential reaches a threshold ϑ, the internal neuron state variable $x_j(t)$ are determined by the impinging spikes, which can be described as the weighted sum of the pre-synaptic contributions:

$$x_j(t) = \sum_{i \in \Gamma_j} \sum_{k=1}^{m} w_{ij}^k \varepsilon(t - (t_i + d^k)) \tag{4}$$

where Γ_j denotes the set of pre-synaptic neurons for the neuron j, m is the number of synaptic terminals between the neuron i and neuron j , w_{ij}^k is the weight of the kth synaptic terminal between the neuron i and neuron j , and t_i is the firing time of the pre-synaptic neuron i.

III. EXPERIMENTS AND RESULTS

A. Clustering of Nonlinearly Separable Ring Dataset

In this example, we consider the ring data consisting of data points in 2-dimensional space as shown in Fig.4(a). The data points in the inner cluster are centered at origin and generated randomly 100 points with the radiuses less than or equal to 0.3. The 100 points generated randomly in the outer cluster form an annular ring at the range from 0.7 to 0.9. It is seen that the

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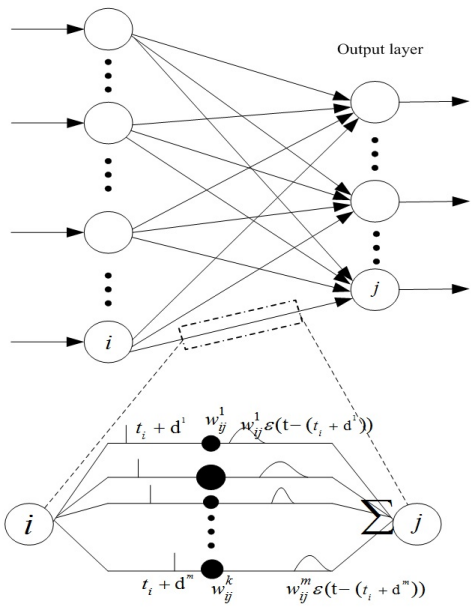


Fig. 3. Architecture of a spiking neural network.

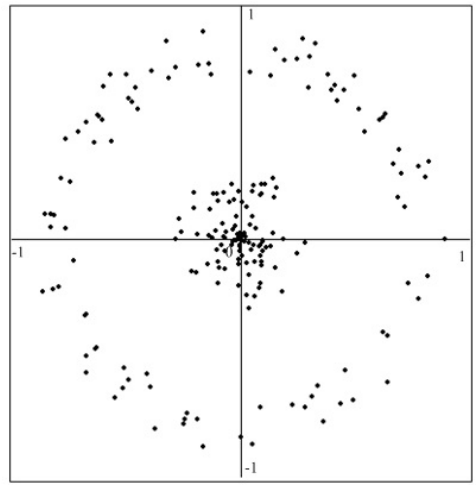
inner and outer clusters of the ring data can be separated by a circular shaped curve. A two layered fully connected feed-forward SNN with ten input neurons and two output neurons was employed in this case. The data is encoded with 5 GF along the x-axis and 5 GF along the y-axis. The parameter values for the simplified STDP are set as the same as in the clustering of linearly separable dataset.

We choose to use 75% of the data for learning and all the data will be in the testing. 10 trials were conducted totally and the mean accuracy is 99.9%. The mean recognition number of points is 197.9 which less than the number of samples 200 because of some input neurons without response in one run (50 ms). Fig.4(b)shows one of the results, the recognition number of points is 199 and the accuracy is 100% (all excitation neurons were clustered correctly). We consider this example of nonlinearly separable data to demonstrate the capabilities of the SNNs in clustering nonlinearly separable data.

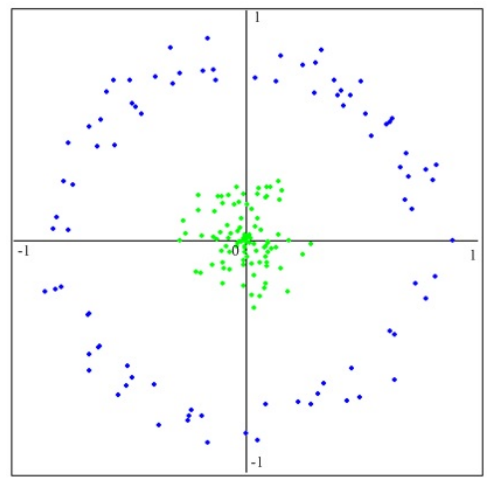
B. Clustering of Fisher Iris Dataset

The Fisher iris species classification problem consists of four flower features (petal width, petal length, sepal width, sepal length) and three classes (three species of the iris plant: Versicolor, Virginica, and Setosa). The dataset consists of 150 samples, 50 for each species. Fig.5 illustrates the data set by means of a two-dimensional projection. In the picture, it is clearly visible how the data for the Iris setosa form a clearly distinct cluster at the right bottom, whereas the clusters of Iris virginica and Iris versicolor overlap and cannot be separated linearly in the projections.

A two layered fully connected feed-forward SNN with 20 input neurons and 3 output neurons were employed in this case. The data is encoded with 5 GF for each input feature. Three output neurons indicate 3 species of the iris plant. The centre and width of each Gaussians neuron was calculated



(a) Two-dimensional ring dataset with 200 points



(b) One of clustering results with different color

Fig. 4. Clustering of nonlinearly separable ring dataset

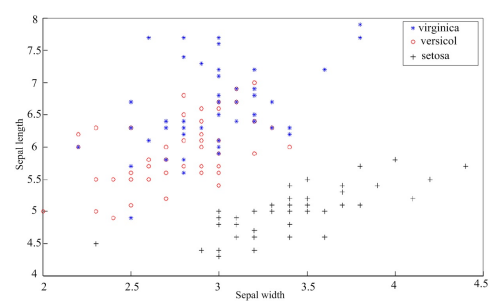


Fig. 5. Dataset. The coordinates are sepal width and sepal length.

using Eq.(5) and Eq.(6). The settings are summarized in Table 2.

The center of the Ith GF is set to

$$\mu = I_{min} + [(2i-3)/2][(I_{max} - I_{min})/(M-2)] \qquad (5)$$

where $i \in 1, 2, \ldots, M$.

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TABLE II
GAUSSIANS NEURON SETTINGS

Gaussians neurons	5
I_{min}	0
I_{max}	50
γ	1.5
Threshold fire line	0.1

The width of the Ith GF is set to

$$\sigma = (1/\gamma)[(I_{max} - I_{min})/(M-2)] \tag{6}$$

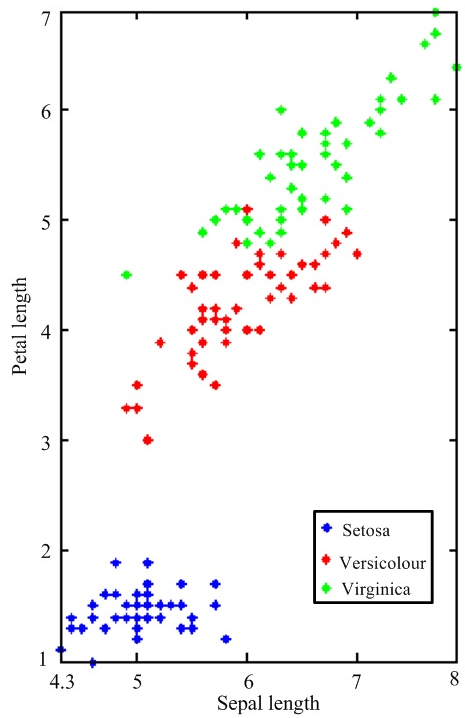
where γ is an adjustment factor and controls the extent of overlap between the GFs.

For this experiment, the parameter values for the simplified STDP are set to: the minimum value for connection strength $w_{min} = 0$, the maximum value for any connection strength $w_{max} = 0.2$, the slope of line in Simplified STDP learning curve $r = 0.2$, the learning rate parameter $\eta = 0.35$. We choose to use 75% of the data for learning and all the data will be in the testing. Fig.6 (a) illustrates the scatter plot by projecting features 1 and features 3 of the dataset onto a 2D plane. Fig.6 (b) illustrates the results of clustering associated with features 1 and features 3.

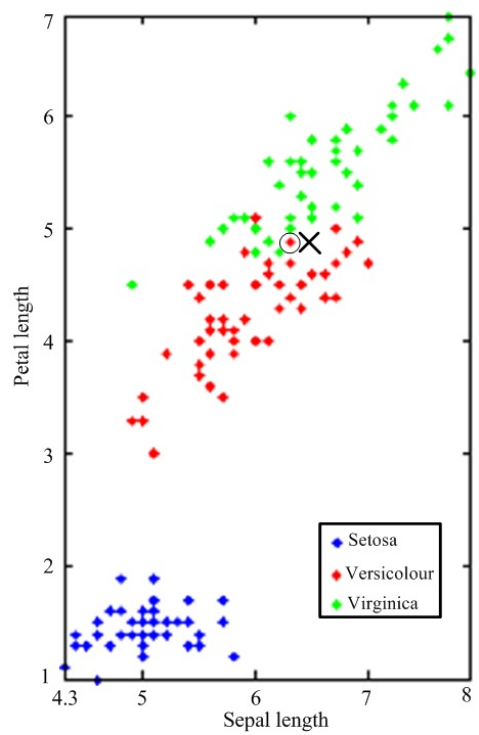
Here, we use only features 1 and features 3 visualize data points scattered in a 2D plane. In fact, we use all the 4 features in the testing. In this experiment, the number of samples is 150, the recognition number of points is 150 and the best accuracy is 95.333%. In [1], Bohte et al. reported their classification yielded 92.6%±0.9% correct classification for the Iris data set, where the inputs were encoded with 4x8 input neurons. Alternative clustering methods, like k-means (k=3) and a self-organizing map (SOM)(with 3 output neurons) yielded somewhat worse results with the accuracy 88.6%±0.1% and 85.33%±0.1% respectively. This result indicates that the clustering capability of our method is at least competitive with similar methods. Also, the model was able to obtain good clustering accuracy with relatively smaller networks. In addition, the population encoding scheme is flexible with varying receptive field widths and the number of neurons.

IV. CONCLUSION

In this paper, we have proposed a method in which a model of STDP learning rule was simplified and used in artificial network based on spiking neurons. Population Encoder Algorithm was utilized to convert the continuous real-number signals to discrete input spike times. Subsequently, the generated spikes are submitted into a two-layered fully connected feed-forward structure SNN which employed Leaky Integrate and Fire (LIF) neurons as nodes and trained by simplified STDP learning rule. Using this method, it is possible to cluster nonlinearly separable data like the ring data. Furthermore, the method was benchmarked using 4-Dimension Fisher iris plant classification problem. The results highlight the algorithm's capability to classify these datasets and can achieve the best accuracy



(a) The Iris dataset with features 1 and 3.



(b) Clustering results with features 1 and 3.

Fig. 6. Clustering of Iris dataset

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of 100% and 95.33% respectively, which demonstrates that the proposed simplified model of STDP learning rule is a viable training algorithm and the SNN approach is suitable for unsupervised clustering.

However, only an SNN with two layers has been considered in this work. For higher dimensional data and hierarchical clustering, it is necessary to consider a multi-layer SNN. In future work we will focus on how to improve the efficiency of the learning method and also explore the hierarchical clustering in a multi-layer SNN.

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